INCH-POUND

MIL-P-70525 (AR)
AMENDMENT 4
18 MARCH 1993

SUPERSEDING
AMENDMENT 3
29 May 1991

MILITARY SPECIFICATION

PROJECTILE, 105MM, HE, RA, XM913/XM927
MOTOR BODY BONDING ASSEMBLY FOR

This amendment forms a part of Military Specification MIL-P-70525 (AR), dated 29 October 1990, and is approved for use by the U.S. Army Armament, Munitions and Chemical Command and is available for use by all Departments and Agencies of the Department of Defense.

PAGE 1

* Title: Change "XM913" to "M913" (and wherever is appears). Change "Projectile" to "Cartridge".

PAGE 4

* 3.2 Delete "9391136" and substitute "9381136".

* 3.3.2 Capitalize "there".

3.3.3 Change "Cracks" to "Discontinuities".

3.3.5.1 Delete "Cracks" in two places and substitute "Discontinuities".

PAGE 5

* 3.3.5.3 Delete "MIL-R-12562" and substitute "MIL-R-13562".

PAGE 9

* 3.7 Delete "4.5.6" and substitute "4.5.8".

AMSC N/A

FSC 1119

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* 3.8.2 Delete "(See Figure 1)" and substitute "(See Figure 3)".

PAGE 10

* Add New Figure 3. X-ray of nozzle (included).

PAGE 12

* 4.3.1.1 Change "Finished Machined Motor Bodies, Unpainted and Unknurled" to "Motor body, Prior to Protective Coating".

PAGE 14

* Finished Machined Motor Bodies Unpainted and Unknurled, Mechanical properties, Delete "4.4.3.2/4.5.5" and substitute "4.4.3.1/4.5.7".

* Nozzle Inserts, Compression test, Delete "3.6.1" and substitute "3.8.1".

* Radiographic Inspection, Delete "3.6.2" and substitute "3.8.2".

PAGE 17

4.4.2.1 Title, delete in its entirety and substitute: "Motor Body, Prior to Weld".

PAGE 18

* 4.4.2.2 Delete title and substitute "Motor Body, prior to protective coating".

Add the following Special defects:

"2	Ultrasonic C-Scan test	100%	3.6	4.5.4
3	Ultrasonic C-Scan test stamp missing	100%	3.2	Visual".

* Major 107 add "(if applicable)".

PAGE 20

* 4.4.2.2 Major 134, Delete in its entirety and add new Minor 211: "211 Radius at cavity base Level V 3.2 4.4.3.7/ Gage".

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X-RAY INSPECTION ZONES

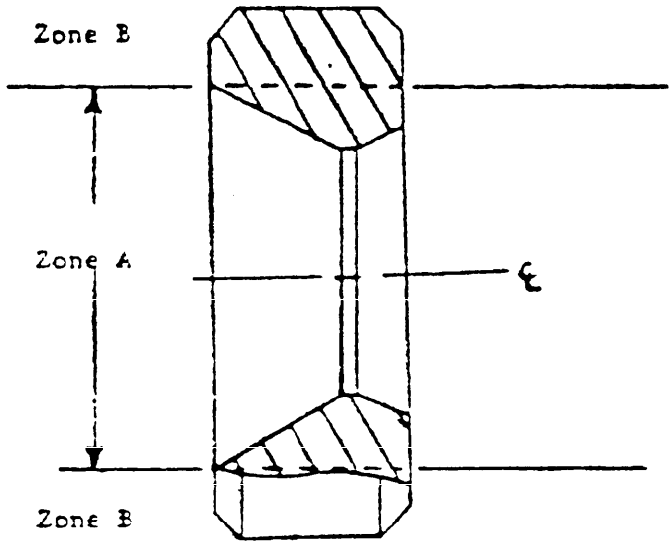


Figure 3

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Major 142, Inspection Methods, delete, "4.5.6" and substitute "4.5.7".

Major 143, Inspection Methods, delete "4.5.2" and substitute "4.5.5".

PAGE 22

* Delete Paragraph "4.4.2.2" and substitute "4.4.2.3".

* Majors 102 and 103, delete in their entirety and substitute the following:

"102	Salt spray test	4.4.3.6.1	3.2	4.5.5.1
103	Protective coating and thickness	4.4.3.6.2	3.2	4.5.5.2".

PAGE 23

* 4.4.2.4 Majors 101, 102, 103, delete in their entirety and substitute the following:

"101	Radiographic inspection	100%	3.8.2	4.5.3.2
102	Grain flow	4.4.3.5.2	3.2	4.4.3.4.2
103	Compression test	Level III	3.8.1	4.4.3.5.1/ 4.5.3.1".

Major 105, delete "(2 Places)".

Add Major 111: "Angle, aft end Level III 3.2 Gage".

PAGE 24

* 4.4.2.5 Major 101, Delete in its entirety and replace with:

"Adhesive not present on forward O.D. nozzle edge or rear nozzle I.D. edge (360 degrees coverage required)	100%	3.2	Visual"

Major 102, delete in its entirety and replace with:

"Gap between aft end of nozzle and motor body	Level III	3.2	Gage"

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PAGE 26

* 4.4.3.2.1 Delete the second and third sentence:

"The chips for the sample . . . selected and dissolved." and substitute:

"The chips for the sample shall consist of the copper overlay between 4.20-.01 and 4.223-.01 inches diameter (full width). All of the chips shall be collected, crushed, and dissolved (see 6.8.1)."

In the sixth sentence, delete "samplie" and substitute "sample".

PAGE 30

* 4.4.3.8.1 Delete "4.5.9" and substitute "4.5.8".

* 4.4.3.8.2 Before last sentence add: "If the lot is rejected, the acceptance criteria reverts back to the beginning of the paragraph".

Delete "4.5.9" and substitute "4.5.8".

PAGE 33

* 4.5.4.1 Second paragraph, first sentence delete "reviously" and substitute "previously".

PAGE 34

* 4.5.5.1 Delete "4.4.3.5.1" at end of paragraph and substitute "4.4.3.6.1".

* 4.5.8 Delete in its entirety and replace with new paragraph 4.5.8:

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"4.5.8 Proving Ground. Ballistic testing shall be performed at Government proving ground in accordance with the applicable Acceptance Test Procedure. The sample motor body assemblies selected in accordance with 4.4.3.8 shall be loaded per 4.5.8.1, and assembled and fired at ambient temperature in an M119 series cannon using a PXR229 propelling charge. The cartridge (projectile, propelling charge, and fuze) shall be uniformly conditioned to the firing temperature prior to testing. Photographic and instrumental coverage shall be recorded for compliance with 3.7. Range and deflections shall be recorded for informational purposes only. In the event of any abnormal condition, as observed by personnel or instrumentation, an attempt shall be made to recover and examine the metal parts for informational use only.

4.5.8.1 Loading Requirements. The projectile for ballistic testing shall be comprised of an inert warhead, and an inert rocket motor. The choice of live/inert delay assemblies and live/inert fuzes is not a requirement of this specification. Use of live rocket motors and live delay assemblies for concurrent testing of rocket motors and/or delays is permitted only with permission from the technical agency."

PAGE 35

* 4.5.9 Delete in its entirety without substitution.

5.1 Add the following:

"... with MIL-STD-1169, Level C. Appropriate cushioning and partitions shall be provided to protect the items from contacting each other during shipment. Sufficient fillers must be used to insure a tight pack. Adequate corrosion protection shall be provided, if required."

* 6.0 NOTES. Add the following parenthetical note:

"(This section contains information of a general or explanatory nature that may be helpful, but is not mandatory.)"

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* 6.2 Acquisition requirements. Delete in its entirety and substitute the following new paragraph:

"6.2 Acquisition requirements. Acquisition documents must specify the following:

a. Title, number, and date of the specification.

b. Issue of DODISS to be cited in the solicitation, and if required, the specific issue of individual documents referenced (see 2.1.1)."

* 6.3 Delete in its entirety and replace with new:

"6.3 Consideration of data requirements. The following data requirements should be considered when this specification is applied on a contract. The applicable Data Item Descriptions (DID's) should be reviewed in conjunction with the specific acquisition to ensure that only essential data are requested/provided and that the DID's are tailored to reflect the requirements of the specific acquisition. To ensure correct contractual application of the data requirements, a Contract Data Requirements List (DD Form 1423) must be prepared to obtain the data, except where DOD FAR Supplement 27.475-1 exempts the requirements for a DD Form 1423.

Reference Paragraph	DID Number	DID Title	Suggested Tailoring
4.4.4	OT-90-12009	AIE Design Documentation	None

The above DID's were those cleared as of the date of this specification. The current issue of DOD 5010.12L, Acquisition Management Systems and Data Requirements Control List (AMSDL), must be researched to ensure that only current, cleared DID's are cited on the DD Form 1423."

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6.4 Delete in its entirety and replace with new:

"6.4 Submission of inspection equipment for design approvals. Gage designs as required shall be submitted to: Commander, ARDEC, ATTN: SMCAR-QAR-I, Picatinny Arsenal, New Jersey 07806-5000. This address shall be specified on the Contract Data Requirements List, DD Form 1423 in the contract. Unless otherwise specified, Data Item OT-90-12009 shall apply."

6.8.1 Add the following new paragraph:

"6.8.1 Chemical analysis. An equivalent method may be submitted to the Government for approval."

The margins of this amendment are marked with an asterisk or vertical line to indicate where changes (additions, modifications, corrections, deletions) from the previous amendment were made. This was done as a convenience only and the Government assumes no liability whatsoever for any inaccuracies in these notations. Bidders and contractors are cautioned to evaluate the requirements of this document based on the entire content irrespective of the marginal notations and relationship to the last previous amendment.

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